# Practice workbook · Lobes of the brain and the cerebellum

Structure and function of the frontal, temporal, parietal and occipital lobes and the cerebellum.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-BRN-01

# 1. Question 17 · part a · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part a only.

**17** Sandy has been asked to investigate the function of 'Region G' of the brain.

She decides to give some tests to one of her female patients who has damaged 'Region G' of their brain:

- Colour naming test: The patient is asked to name what colour they see when presented with a coloured piece of paper.
- Recall test: The patient is asked to recall the months of the year in order and when they were born.
- Pressure test: Physical pressure is applied to the patient when they are blindfolded and they have to say where they can feel the pressure being applied.

The patient performed well on the recall test and pressure test but performed poorly on the colour naming test.

(a) Explain which region of the brain the patient is likely to have damaged from their performance in the tests.

You should refer to the function of the damaged region in your answer.

**(2)**

(b) Explain **one** improvement Sandy could have made to her investigation.

**(2)**

**(Total for Question 17 = 4 marks)**

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## 1. Question 17 · part a · 2 marks · Working and answer

Working and answer

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## 2. Question 17 · part whole · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

**17** Malvina was investigating the role of different brain regions.

She organised for three groups of participants to carry out two tasks.

All the participants in group 1 had damage to region X of the brain. All the participants in group 2 had damage to region Y of the brain. A control group (group 3) had no damage to their brain.

All groups were required to carry out two tasks.

- Task 1: To walk along a narrow balance beam without touching the floor below.
- Task 2: To look at 10 images of common household objects and name them.

The average scores of the participants are shown in **Table 2**.

| Group | Task 1: Beam task (number of times they touched the floor) | Task 2: Image task (number of objects correctly named out of 10) |
|---|---|---|
| **Group 1 average score** | 16 | 10 |
| **Group 2 average score** | 1 | 1 |
| **Group 3 average score** | 1 | 10 |

**Table 2**

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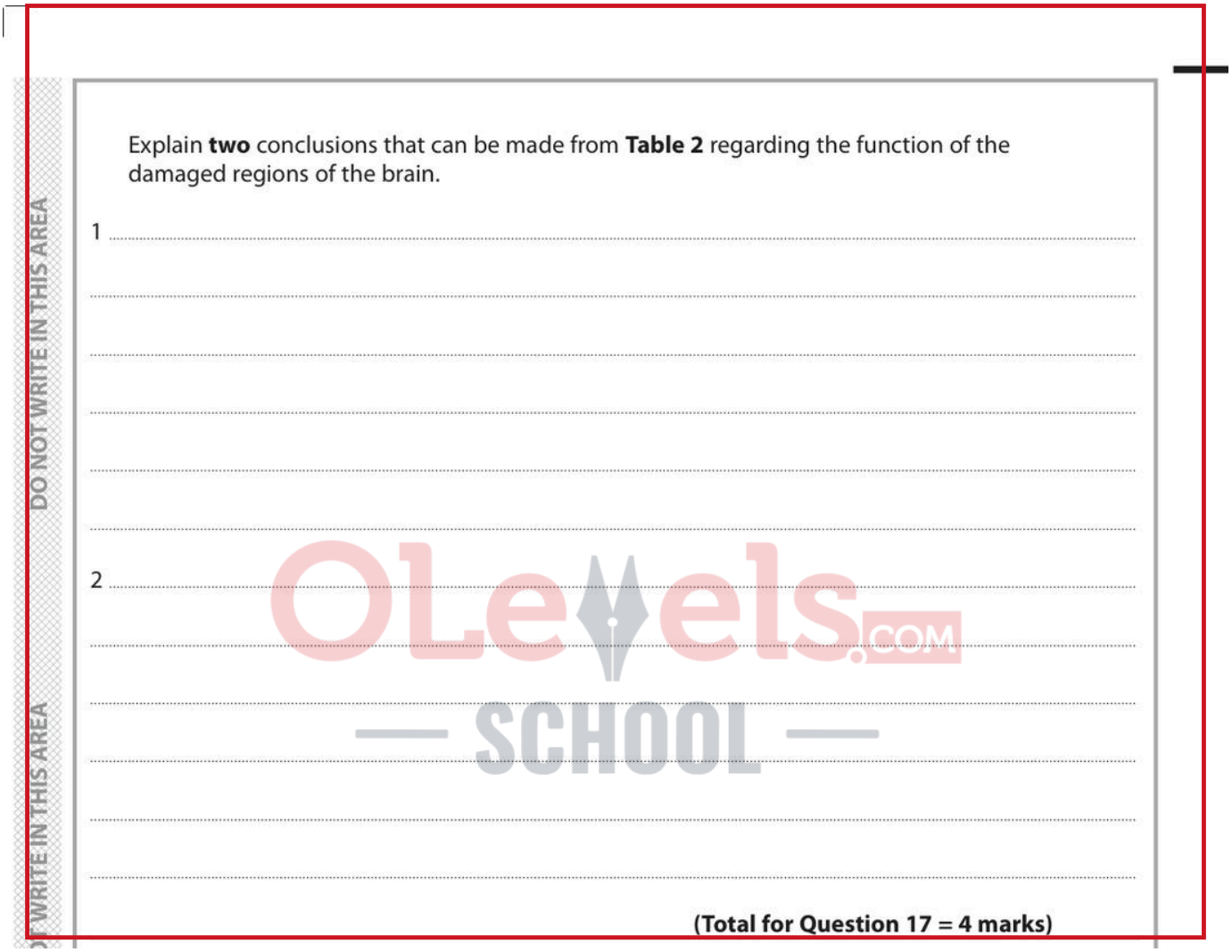


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## 2. Question 17 · part whole · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.



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Explain **two** conclusions that can be made from **Table 2** regarding the function of the damaged regions of the brain.

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**(Total for Question 17 = 4 marks)**

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## 2. Question 17 · part whole · 4 marks · Working and answer

Working and answer

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## 3. Question 20 · part whole · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

**20** Five patients with damage to a region of the brain were investigated by Lynne. The brain lobe damaged is shown in **Figure 4**.

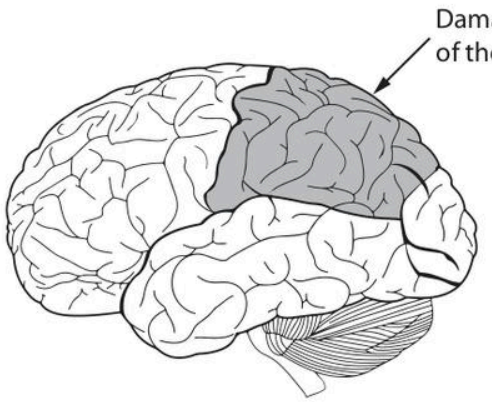



**Figure 4**

She gave them tasks involving problem solving, pain, and memory.

The tasks were:

- Problem solving: solve 10 word-based puzzles where participants were given three words and had to think of a fourth word related to them
- Pain: three pads were placed on the body and the temperature was increased on one pad and participants had to say which of the pads was active
- Memory: read 20 trigrams and try to remember as many as possible for a retrieval test after five minutes.

Explain what Lynne is likely to find with her patients.

You must refer to the function of the damaged brain lobe in your response.

**(Total for Question 20 = 2 marks)**

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## 3. Question 20 · part whole · 2 marks · Working and answer

Working and answer

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## 4. Question 17 · part whole · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

**17 Figure 4** shows an image of a human brain.

Complete the boxes provided with the areas of the brain.

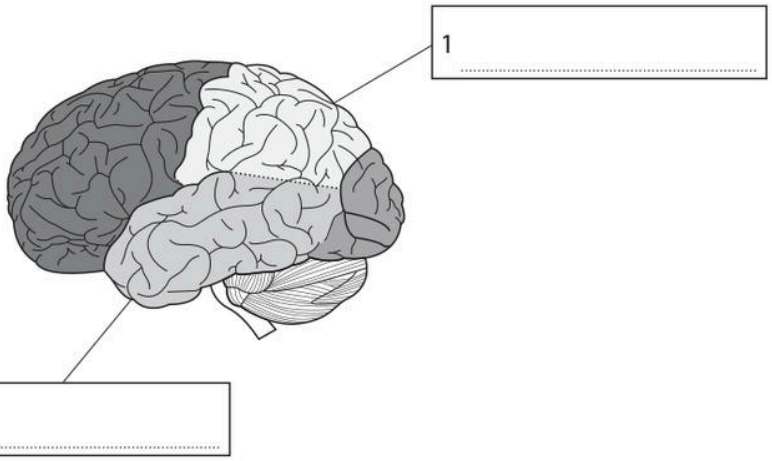



**Figure 4**

**(Total for Question 17 = 2 marks)**

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Working and answer

## 5. Question 16 · part whole · 2 marks

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Further practice · Attempt part whole only.

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**16** Diogo is investigating the function of the temporal lobes. He recruits participants with and without damage to their temporal lobes to participate in his study. Diogo believes that he will better understand the function of the temporal lobes by seeing if there is any difference in the participant performance on a task.

Diogo asks his participants to listen to a simple song that he has created. He then asks them to try to sing the song out loud.

Explain what Diogo is likely to find in his study.

You should refer to the function of the temporal lobes in your answer.

**(Total for Question 16 = 2 marks)**

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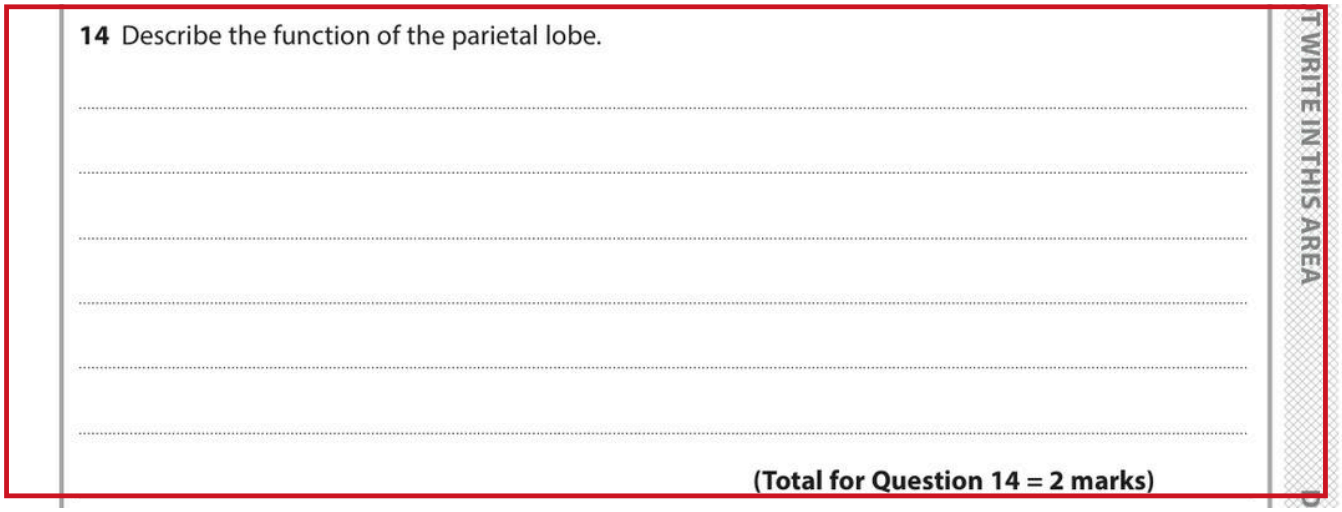
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Working and answer

## 6. Question 14 · part whole · 2 marks

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Further practice · Attempt part whole only.


**14** Describe the function of the parietal lobe.

(Total for Question 14 = 2 marks)

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Working and answer

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